# Project Report

| | |
|---|---|
| **Date** | **28-06-2026** |
| **Project Name** | **Automated Network Request Management** |
| **Team ID** | |
| **Maximum Marks** | |

## Project Title: Automated Network Request Management in ServiceNow

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### Project Overview

Automated Network Request Management in ServiceNow is designed to simplify and streamline the process of handling network-related service requests through automation. Organizations frequently receive requests for network access, device configuration, connectivity issues, IP allocation, VPN access, and other infrastructure services. Traditional manual processing often results in delays, inconsistent approvals, and limited visibility into request status.

This project leverages the capabilities of the ServiceNow platform to automate the complete request lifecycle. Users can submit requests through a self-service Service Catalog, after which automated workflows validate the request, trigger approval processes, create records, send notifications, and update request statuses without requiring extensive manual intervention.

By integrating Service Catalog, Catalog Variables, UI Policies, Custom Tables, Flow Designer, Approval Management, and Email Notifications, the system provides a scalable and efficient solution that improves operational efficiency, enhances user experience, ensures compliance with organizational policies, and maintains complete audit trails for every network request.

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## 1. Introduction

Modern organizations depend heavily on reliable network infrastructure for daily business operations. Employees regularly submit requests for network connectivity, VPN access, IP allocation, router configuration, switch configuration, network relocation, and device onboarding. Managing these requests manually consumes significant administrative effort and increases the likelihood of delays and errors.

As organizations continue to grow, the volume of network-related requests also increases. Manual request handling makes it difficult to maintain Service Level Agreements (SLAs), monitor request progress, and ensure consistent approval procedures.

The **Automated Network Request Management in ServiceNow** project addresses these challenges by implementing a centralized, automated request management system. The solution enables end users to submit requests through a user-friendly Service Catalog while ServiceNow automates request validation, approval routing, record creation, email notifications, and request tracking.

The system provides:

- Faster request processing
- Standardized workflows
- Reduced manual effort
- Improved transparency
- Better compliance
- Accurate request tracking
- Improved user satisfaction

Using ServiceNow's low-code capabilities, the project demonstrates how IT Service Management (ITSM) can automate repetitive operational tasks while maintaining flexibility for future enhancements.

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## 2. Project Objectives

The primary objective of this project is to design and implement an automated network request management system within the ServiceNow platform that minimizes manual intervention and accelerates request fulfillment.

The project aims to provide a structured environment where users can easily submit network requests through a Service Catalog while the platform automatically handles approvals, notifications, record creation, and workflow execution.

Specific objectives include:

- Develop a centralized Service Catalog for network-related requests.
- Design dynamic request forms using Catalog Variables.
- Implement UI Policies to display fields dynamically based on user selections.
- Create custom tables for storing network request information.
- Configure Flow Designer to automate request processing.
- Automate approval workflows for managers or network administrators.
- Send automated email notifications during different stages of request processing.
- Maintain complete audit records for all network requests.
- Improve request turnaround time and SLA compliance.
- Build a scalable solution that can support future network services.

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## 3. Key Features

The proposed system includes the following major features:

- User-friendly Service Catalog for submitting network requests.
- Dynamic forms with intelligent field visibility.
- Catalog Variables for capturing request information.
- Variable Sets for reusable user information.
- Automated approval workflows using Flow Designer.
- Custom Network Database table for request storage.
- Automatic creation of backend records.
- Email notifications during approval and completion stages.
- Real-time request status tracking.
- Audit-ready approval history.
- Scalable workflow architecture.
- Reduced manual processing time.
- Better SLA compliance.

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## 4. Project Milestones

The project implementation follows five major milestones.

### Milestone 1: Catalog Creation

**Objective**

Establish a structured Service Catalog for network request management.

Activities:

- Identify different types of network requests.
- Create Service Catalog.
- Create Catalog Categories.
- Create Catalog Items.
- Provide meaningful names and descriptions.
- Organize request taxonomy according to ITSM standards.

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### Milestone 2: Form Setup

**Objective**

Design dynamic forms to collect all necessary request information.

Activities:

- Create Catalog Variables.
- Configure Variable Sets.
- Implement Catalog UI Policies.
- Configure mandatory fields.
- Apply field validation.
- Enable dynamic show/hide functionality.

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### Milestone 3: Approval Integration

**Objective**

Automate approval workflows before request fulfillment.

Activities:

- Configure approval hierarchy.
- Create Flow Designer approval actions.
- Implement manager approvals.
- Enable email notifications.
- Maintain approval audit logs.

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### Milestone 4: Testing

**Objective**

Validate the complete automation workflow.

Activities:

- Unit testing.
- Catalog testing.
- Flow testing.
- Approval testing.
- Notification testing.
- End-to-end request execution.
- User Acceptance Testing (UAT).

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### Milestone 5: Deployment

**Objective**

Deploy the application for organizational use.

Activities:

- Validate configurations.
- Publish catalog items.
- Monitor workflow execution.
- Train end users.
- Monitor post-deployment performance.

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## 5. ServiceNow Developer Setup

Before developing the application, a ServiceNow Personal Developer Instance (PDI) must be created. The PDI provides a fully functional ServiceNow environment where applications can be developed, tested, and deployed without affecting production systems.

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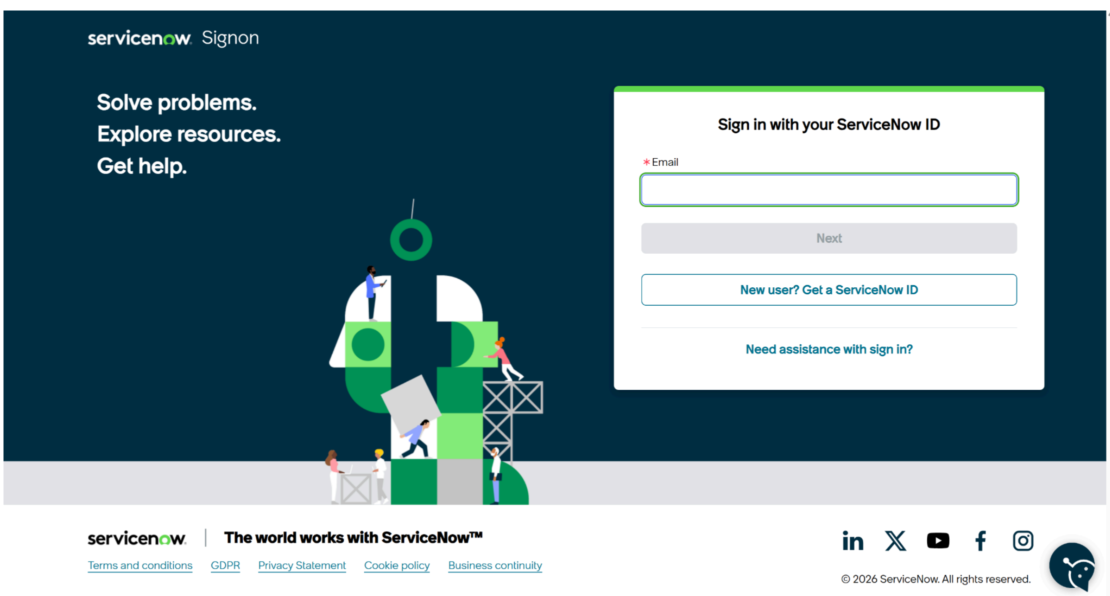
## 5.1 Creating a ServiceNow Developer Account

To begin development, perform the following steps:

1. Visit the ServiceNow Developer Portal.
2. Click **Sign Up** to create a free developer account.
3. Enter the required information:
   - First Name
   - Last Name
   - Email Address
   - Country
   - Password
4. Verify the email address using the activation link sent to your registered email.
5. Log in to the ServiceNow Developer Portal.
6. Click **Start Building**.
7. Request a **Personal Developer Instance (PDI)**.
8. Wait for the instance to be provisioned.
9. Launch the instance and log in using the generated credentials.

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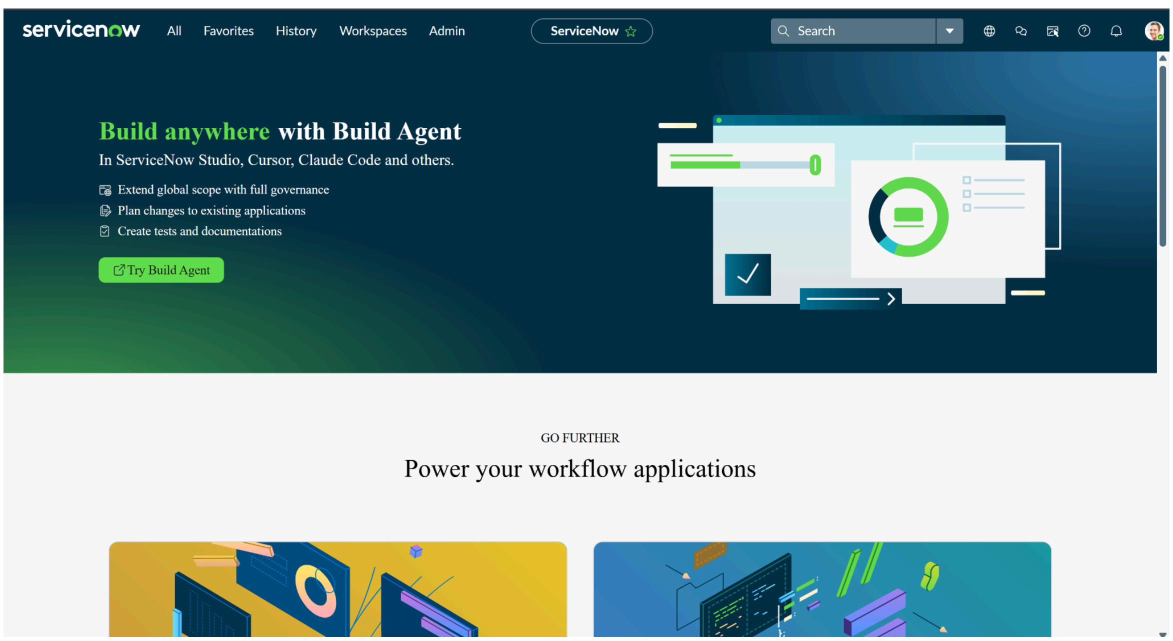
## 5.2 Accessing Creator Studio

Once the Personal Developer Instance is ready, ServiceNow redirects the developer to Creator Studio.

Creator Studio provides a no-code and low-code environment for building applications. It simplifies application development by allowing developers to create forms, workflows, tables, automation, and integrations through an intuitive graphical interface.

Creator Studio offers features such as:

- Application Creation
- Data Modeling
- Flow Automation
- Form Builder
- Catalog Item Development
- Workflow Configuration

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## 6. Project Implementation in ServiceNow

The implementation of the project consists of multiple modules. Each module contributes to automating the complete network request lifecycle.

The implementation includes:

- Service Catalog Creation
- Catalog Variables Configuration
- Variable Sets
- Catalog UI Policies
- Custom Table Creation
- Field Configuration
- Related Lists
- Flow Designer
- Approval Integration
- Email Notifications
- Record Updates

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## 6.1 Service Catalog Creation

The Service Catalog provides a self-service interface where users can submit network-related requests.

The Network Request catalog item was created to allow users to request various network services from a single interface.

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### Steps to Create Service Catalog

1. Open the **Application Navigator**.
2. Search for **Service Catalog**.
3. Navigate to:

**Service Catalog → Maintain Items**

4. Click **New**.
5. Enter the following details:

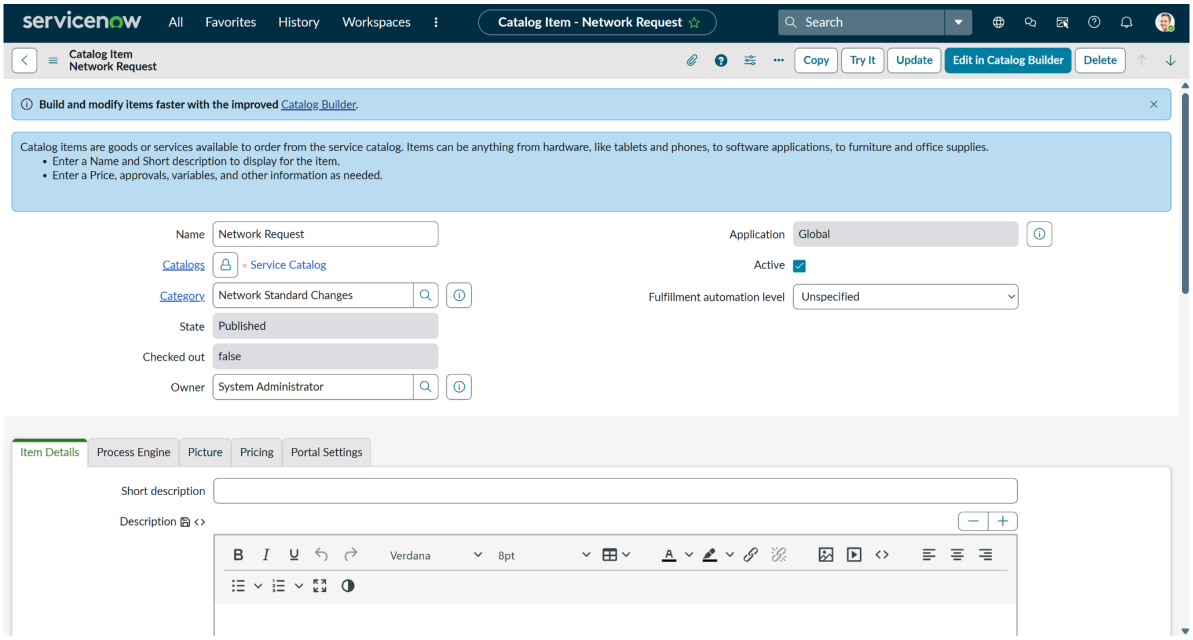
   **Catalog Item Name:** Network Request

   **Catalog:** Service Catalog

   **Category:** Network

   **Short Description:** Automated Network Request Management

6. Save the Catalog Item.

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## Purpose of the Catalog

The Service Catalog serves as the entry point for all network requests.

Users can submit requests such as:

- New Network Connection
- Network Relocation
- Device Registration
- Network Configuration
- Connectivity Support
- Device Information Submission

Using the catalog ensures standardized request submission and eliminates inconsistent manual emails or phone requests.

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## 6.2 Catalog Variables Configuration

Catalog Variables are used to collect information from the requester.

These variables are displayed as fields on the Service Catalog form.

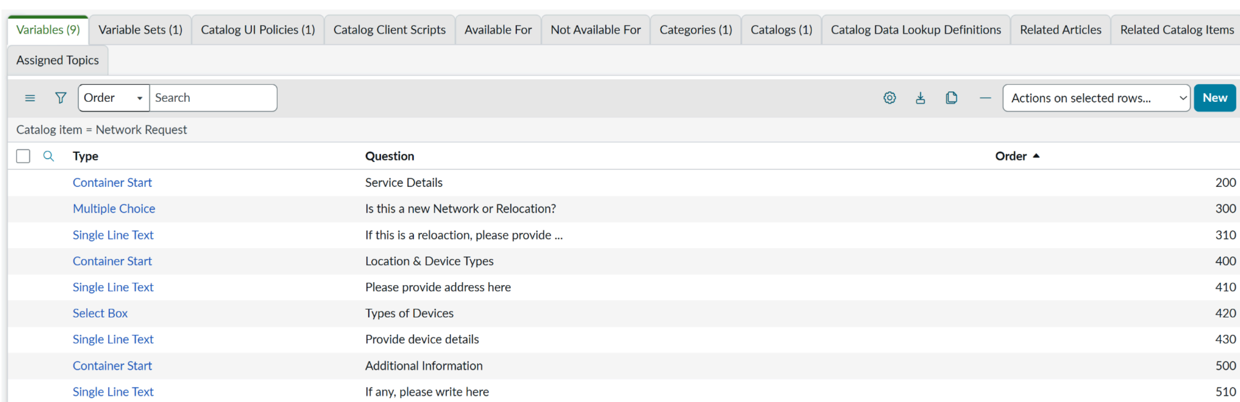
Each variable captures specific information that is later used during workflow automation.

## Creating Catalog Variables

Perform the following steps:

1. Open the **Network Request** Catalog Item.
2. Scroll to the **Variables** related list.
3. Click **New**.
4. Configure the following properties:
- Variable Type
- Question
- Name
- Order
- Tooltip
- Help Text
- Mandatory Status
5. Save the variable.

Repeat the process for all required variables.




| Type | Question | Order |
|---|---|---|
| Container Start | Service Details | 200 |
| Multiple Choice | Is this a new Network or Relocation? | 300 |
| Single Line Text | If this is a reloaction, please provide ... | 310 |
| Container Start | Location & Device Types | 400 |
| Single Line Text | Please provide address here | 410 |
| Select Box | Types of Devices | 420 |
| Single Line Text | Provide device details | 430 |
| Container Start | Additional Information | 500 |
| Single Line Text | If any, please write here | 510 |

## Variable Configuration

The project uses multiple variable types depending on the information required.

| Variable Name | Type |
|---|---|
| Service Details | Container Start |
| Is this a new Network or Relocation? | Multiple Choice |
| If this is a relocation, please provide yo | Single Line Text |
| Location & Device Types | Container Start |
| Please provide address here | Single Line Text |
| Types of Devices | Select Box |
| Provide device details | Single Line Text |
| Additional Information | Container Start |
| If any, please write here | Single Line Text |

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## Variable Types

### 1. Reference Variable

Used to reference records from another table.

Example:

Opened On Behalf Of → User Table

This enables automatic retrieval of user information.

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**2. Choice Variable**

Displays predefined options.

Example:

Network Request Type

Options:

- New Connection
- Relocation
- Maintenance
- Configuration Change

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**3. Single Line Text**

Captures short information.

Examples:

- Email
- Username
- Address
- Phone Number

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**4. Multi-Line Text**

Used for detailed descriptions.

Example:

Device Details

Purpose:

Allows users to describe the device or issue in detail.

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**5. Attachment Variable**

Allows users to upload supporting documents.

Examples:

- Identity Proof
- Device Purchase Invoice
- Approval Documents

And many more.

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### Auto-Populated Variables

Certain variables are automatically populated using **Dot Walking**.

When the user selects **Opened On Behalf Of**, the following fields are filled automatically:

- Email Address
- Username
- Phone Number

This reduces manual data entry and minimizes errors.

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### Benefits of Catalog Variables

- Standardized data collection
- Improved data accuracy
- Easier automation
- Better reporting
- Reduced manual validation

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## 6.3 Variable Set Configuration

Variable Sets are reusable groups of variables.

Instead of creating common variables repeatedly for multiple catalog items, Variable Sets allow developers to define them once and reuse them wherever required.

Examples of reusable fields include:

- Opened On Behalf Of
- Email
- Username
- Phone Number
- Document Upload

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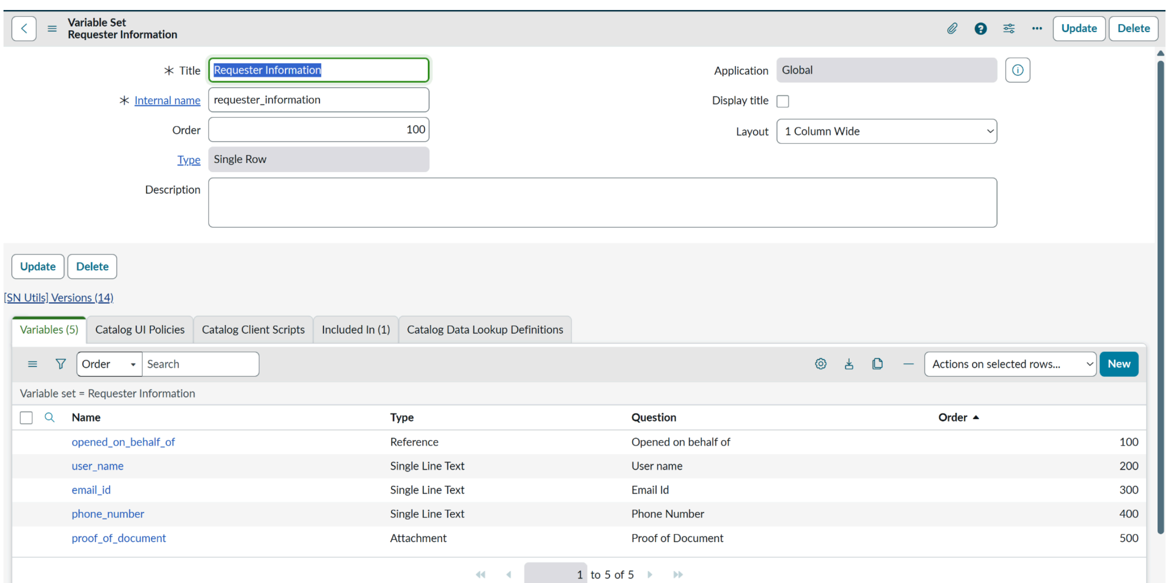
## Creating Variable Sets

1. Navigate to **Service Catalog** → **Variable Sets**.
2. Click **New**.
3. Provide:
- Name
- Description
4. Add all required variables.
5. Save the Variable Set.
6. Associate the Variable Set with the Network Request Catalog Item.

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## Advantages of Variable Sets

- Reduces duplicate configuration
- Easier maintenance
- Consistent user experience
- Reusable across multiple Catalog Items
- Faster development

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## 6.4 Catalog UI Policy Configuration

Catalog UI Policies improve the user experience by dynamically controlling the visibility, mandatory status, and read-only state of form fields based on user input.

In this project, UI Policies are implemented to display only the relevant fields depending on the type of network request selected by the user. This minimizes unnecessary inputs and ensures that only required information is collected.

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### Objective

The primary objective of implementing Catalog UI Policies is to:

- Improve form usability.
- Reduce user confusion.
- Display fields dynamically.
- Enforce mandatory fields only when required.
- Prevent invalid or incomplete submissions.

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### Scenario

If the user selects **Device Type = Others**, an additional field named **"Please Specify Device"** becomes visible.

Similarly, if the user selects **Relocation** as the request type, the **Relocation Address** field is displayed.

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### Steps to Configure Catalog UI Policy

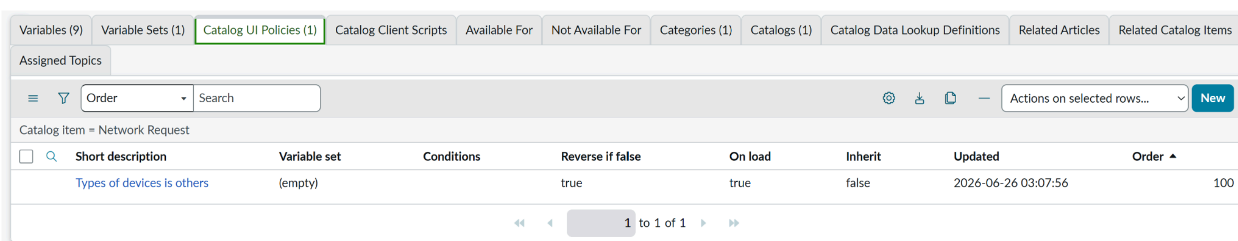
1. Navigate to **Service Catalog → Catalog Definitions → Maintain Items**.

2. Open the **Network Request** Catalog Item.
3. Scroll down to the **Catalog UI Policies** related list.
4. Click **New**.
5. Enter the following information:
    - **Applies To:** Catalog Item
    - **Catalog Item:** Network Request
    - **Description:** Dynamic Field Visibility
6. Configure the condition.

Example:

Device Type is Others

7. Save the UI Policy.
8. Create a **UI Policy Action**.
9. Select the field **Please Specify Device**.
10. Set **Visible = True**.
11. Update the UI Policy.
12. Test the Catalog Item.

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## Benefits of UI Policies

- Dynamic forms
- Better user experience
- Cleaner interface
- Reduced validation errors
- Improved data quality

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## 6.5 Creation of Custom Table

A custom table was created to store all submitted network requests.

Instead of relying only on catalog requests, maintaining a dedicated table allows administrators to manage records efficiently, generate reports, and monitor request progress.

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### Table Details

**Table Name:** Network Database

**Purpose:** Store all submitted network requests.

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### Steps to Create Table

1. Navigate to

**System Definition → Tables**

2. Click **New**.
3. Enter the following information:
- Label
- Table Name
- Application
4. Enable **Auto Generate Schema**.
5. Click **Submit**.

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### Advantages

- Centralized request storage
- Easy reporting
- Better auditing
- Supports future enhancements
- Independent data management

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## 6.6 Field Creation

After creating the table, fields were added to store request information.

Each field corresponds to one Catalog Variable.

## Steps

1. Open the custom table.
2. Navigate to the **Columns** tab.
3. Click **New**.
4. Select the appropriate field type.
5. Enter the Column Label.
6. Provide the Column Name.
7. Save the field.

Repeat the process until all required fields are created.

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## Fields Created

| Field Name | Data Type |
|---|---|
| Request Number | String |
| Requested For | Reference |
| Work Status | Choice |
| Device Type | Choice |
| Device Details | String |
| Customer Address | String |
| Assignment Group | Reference |
| Date of Enquiry | Date |
| Phone Number | String |
| Email ID | String |
| Approval Status | Choice |

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## Purpose of Fields

These fields maintain all necessary information required to process network requests.

The custom table also serves as the backend database for reports and workflow automation.

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## 6.7 Request Approval Related List

Approvals are an essential part of network request management.

To maintain approval history, a Related List was configured between the custom table and the Approval table.

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### Creating Relationship

Navigate to:

**System Definition → Relationships**

Click **New**.

Configure the following:

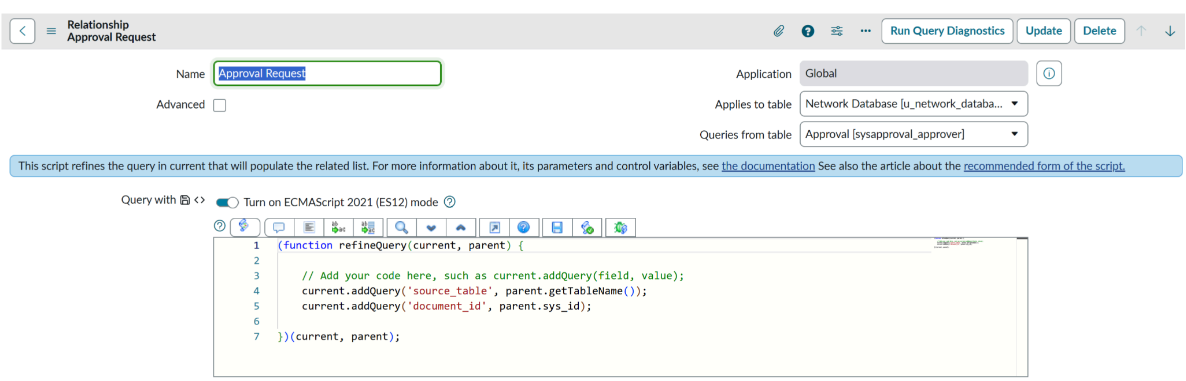
**Relationship Name:** Approval Request

**Applies To:** Network Database

**Queries From:** sysapproval_approver

**Active:** True

Save the relationship.



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### Adding Related List

1. Open Form Designer.
2. Select the Network Database table.
3. Add a Related List.
4. Choose **Approval Requests**.
5. Save the form.

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### Benefits

- Displays approval history.
- Tracks approver decisions.
- Maintains audit records.
- Improves transparency.

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## 6.8 Flow Designer Implementation

Flow Designer is the core automation engine used in this project. It enables the automatic execution of predefined actions whenever a network request is submitted through the Service Catalog. The flow eliminates manual intervention by creating records, requesting approvals, sending notifications, and updating request statuses based on approval outcomes.

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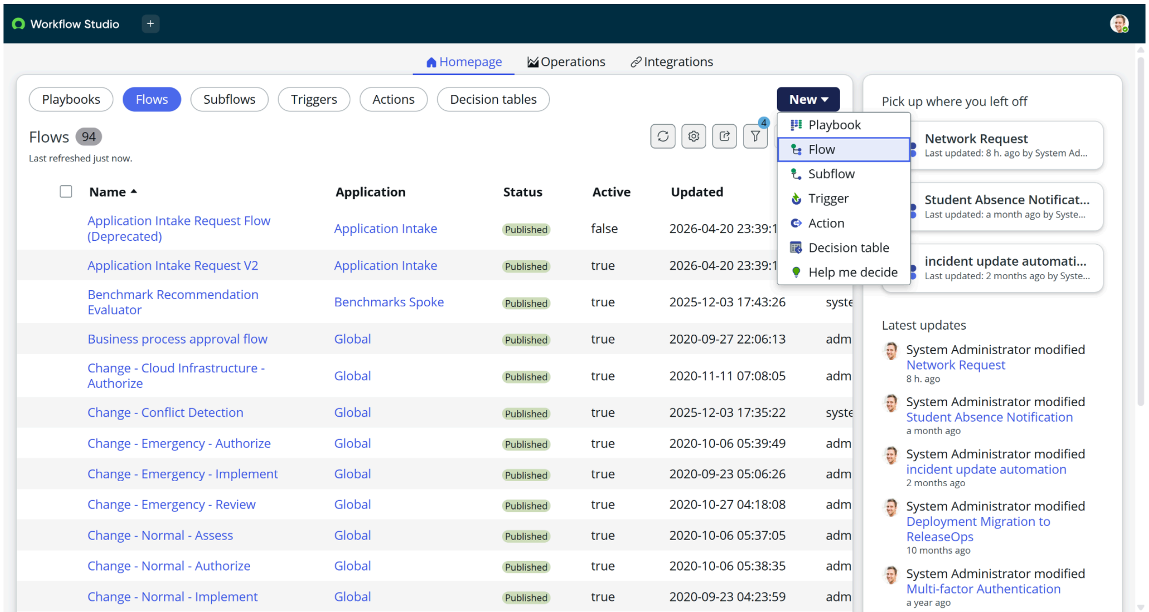
### 6.8.1 Flow Creation

The first step is to create a new flow that automates the complete network request lifecycle.

### Steps to Create the Flow

1. Navigate to **Flow Designer** from the Application Navigator.
2. Click **New**.
3. Select **Flow**.
4. Enter the following details:
    - **Flow Name:** Network Request Management
    - **Description:** Automates the processing of network service requests.
5. Click **Build Flow**.

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### 6.8.2 Configuring the Trigger

The trigger defines when the flow should start.

In this project, the flow is triggered whenever a user submits the **Network Request** catalog item.

## Steps

1. Click the **+** icon.
2. Select **Trigger**.
3. Choose **Service Catalog**.
4. Select **Requested Item Created**.
5. Select the **Network Request** catalog item.
6. Save the trigger.

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### 6.8.3 Get Catalog Variables

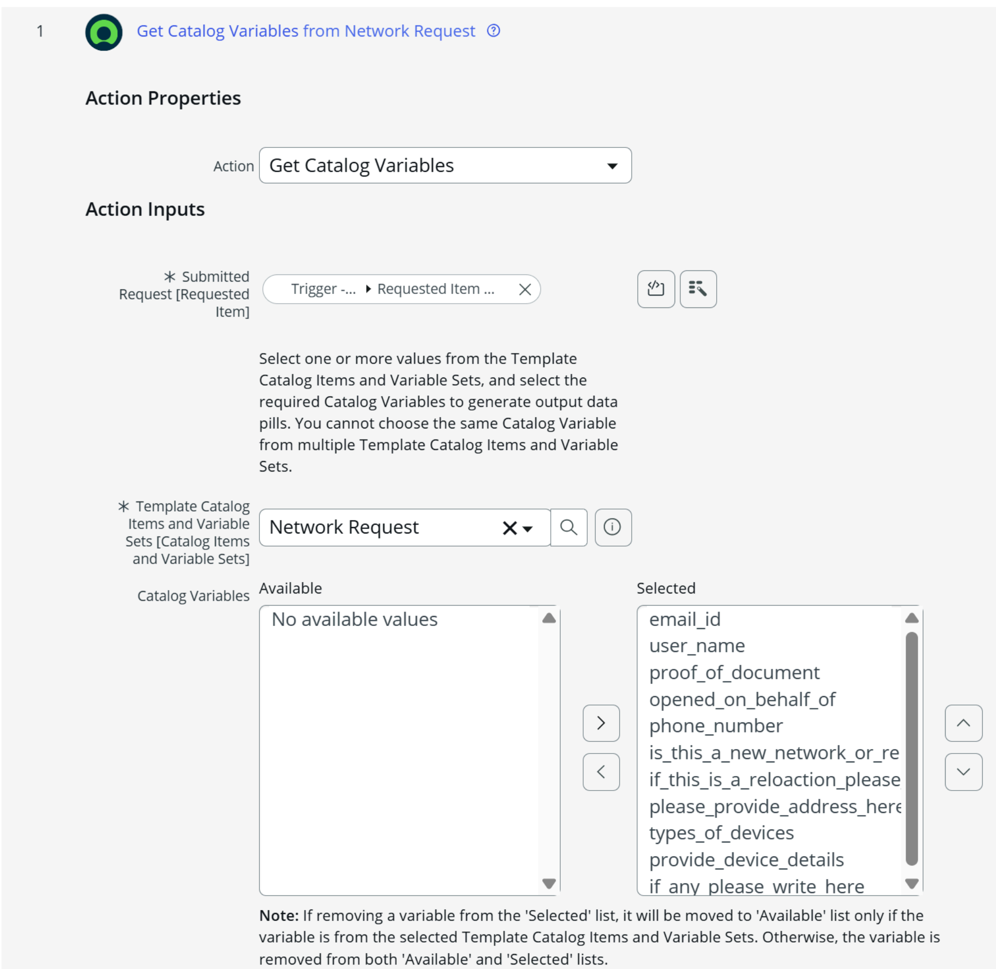
After the flow starts, all information entered by the user is retrieved using the **Get Catalog Variables** action.

## Steps

1. Click **Add Action**.
2. Search for **Get Catalog Variables**.
3. Select the action.
4. Configure the Requested Item as the input.
5. Select all required variables.
6. Save the configuration.

This action retrieves details such as:

- Requested For
- Email ID
- Phone Number
- Device Type
- Device Details
- Address
- Request Type



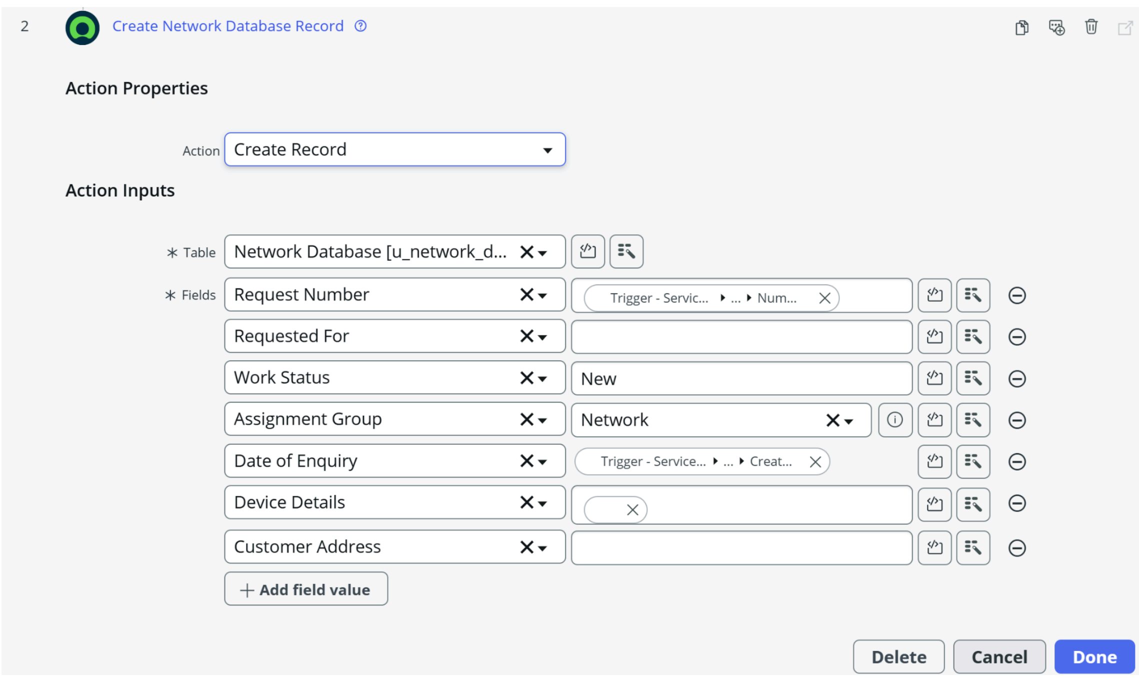
### 6.8.4 Create Record

The retrieved information is stored in the custom **Network Database** table.

## Steps

1. Add the **Create Record** action.
2. Select the **Network Database** table.
3. Map the catalog variables to the corresponding table fields.
4. Save the action.

The record contains all relevant request information, allowing administrators to monitor and process requests efficiently.




### 6.8.5 Approval Process

To ensure that only authorized requests are fulfilled, an approval process is integrated into the workflow.

## Steps

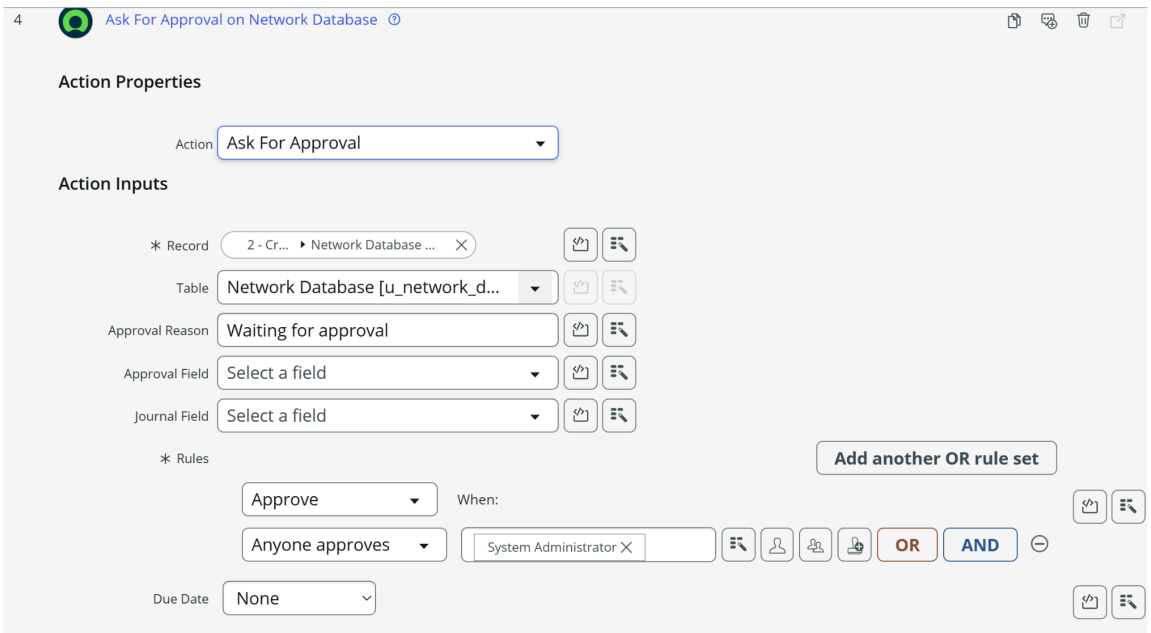
1. Add the **Ask for Approval** action.
2. Select the newly created Network Database record.
3. Configure the approval rule.

4. Specify the approver (Manager or Network Administrator).
5. Set the approval reason.
6. Save the configuration.

Possible approval outcomes include:

- Approved
- Rejected
- Pending

The approval history is automatically maintained in the system.

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### 6.8.6 Flow Logic

Conditional logic is implemented to determine the next action based on the approval result.

## Approved

If the request is approved:

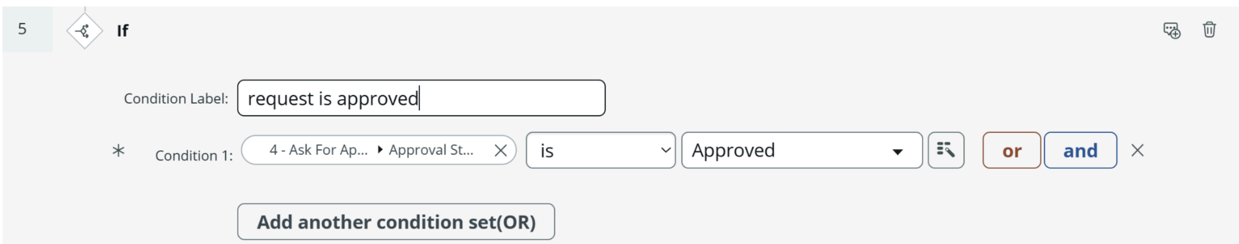
- Send approval email.
- Update the request status to **Approved**.
- Continue with request fulfillment.

## Rejected

If the request is rejected:

- Send rejection email.
- Update the request status to **Rejected**.
- Stop further processing.

This ensures that only approved requests proceed through the workflow.

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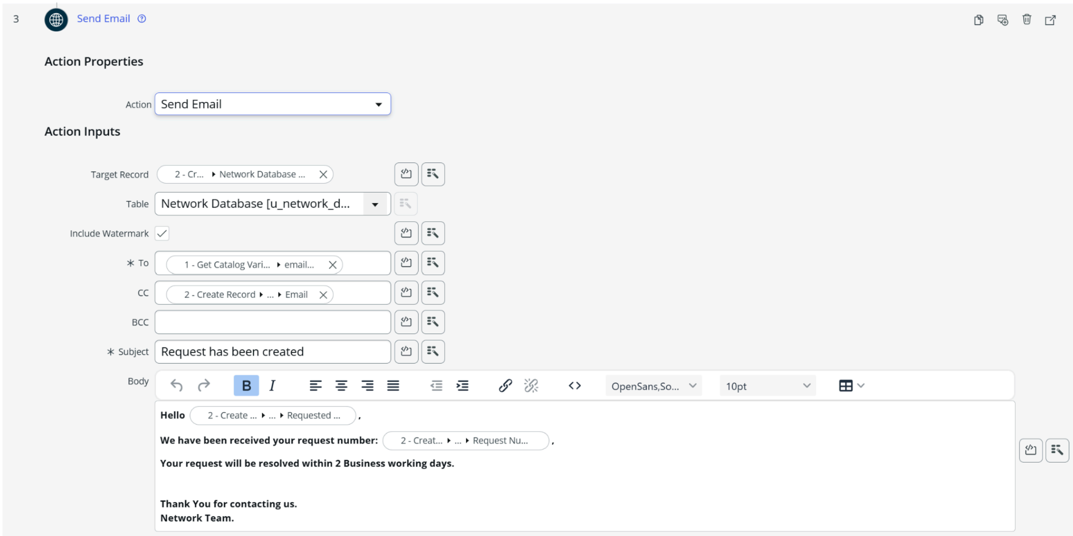
### 6.8.7 Email Notifications

Automatic email notifications keep users informed about the status of their requests.

The system sends notifications at different stages, including:

- Request Submitted
- Approval Pending
- Request Approved
- Request Rejected
- Request Completed

Each email contains the request details and current status, improving communication and transparency.

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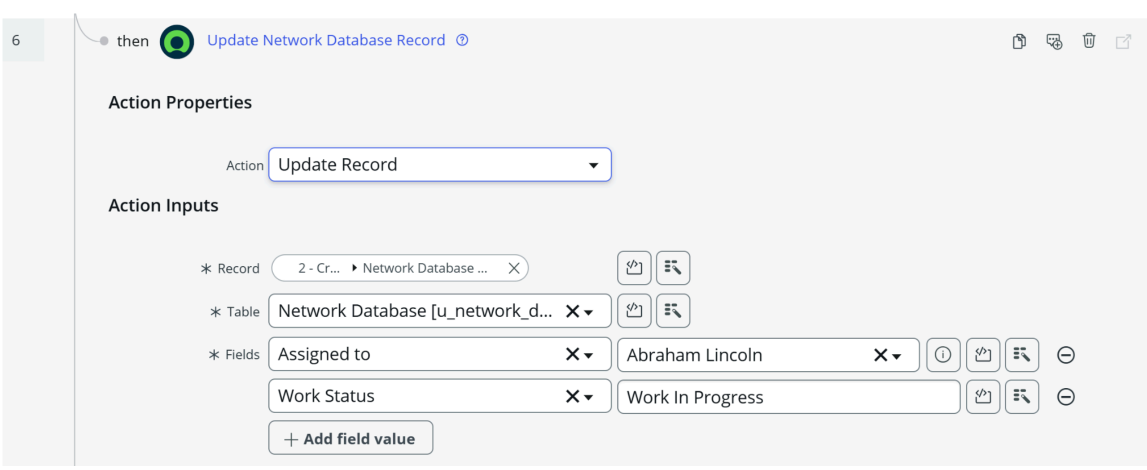
## 6.8.8 Update Record

Once the approval process is complete, the corresponding record in the Network Database table is updated automatically.

The system updates information such as:

- Approval Status
- Work Status
- Completion Date
- Assigned Team

This ensures that the database always reflects the latest status of each request.

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## 6.9 Workflow Summary

The complete workflow of the project is illustrated below.

1. User submits a Network Request through the Service Catalog.
2. Flow Designer is triggered automatically.
3. Catalog Variables are retrieved.
4. A record is created in the Network Database table.
5. Approval request is sent to the designated approver.
6. The approver reviews the request.
7. Based on the approval decision:
    - If Approved, the request proceeds and the user is notified.
    - If Rejected, the request is closed and the user is informed.
8. The record is updated with the latest request status.

# Network Request

Active

TRIGGER

Service Catalog

ACTIONS Select multiple

1 Get Catalog Variables from Network Request

2 Create Network Database Record

3 Send Email

4 Ask For Approval on Network Database

5 **If** request is approved

6 then Update Network Database Record

7 Send Email

8 **End Flow**

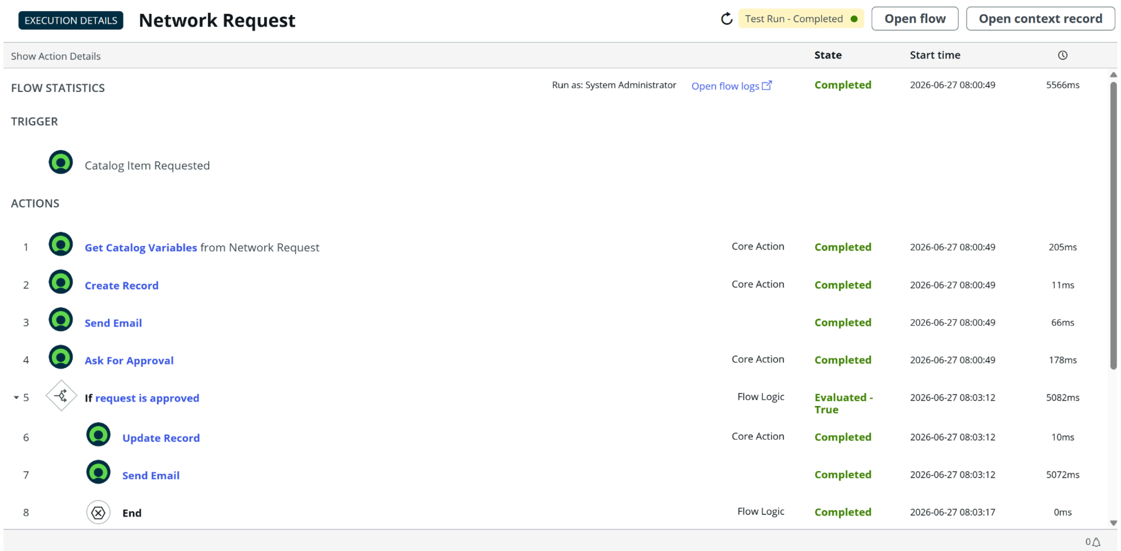
9 **If** request is rejected

10 then Send Email

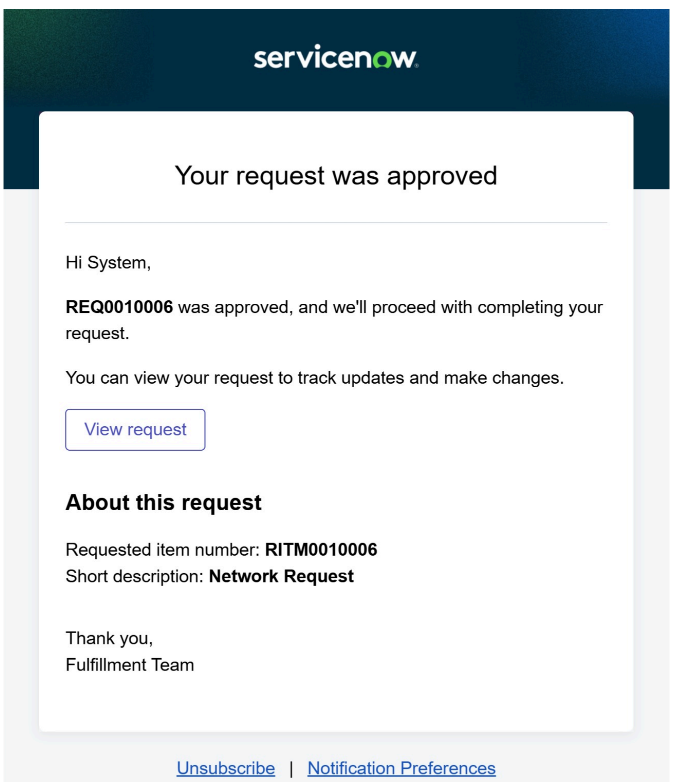
11 **End Flow**

# 7. Screenshots of Output

## 1. Flow Execution Test




## 2. Email Notification



## 8. Conclusion

The **Automated Network Request Management in ServiceNow** project successfully demonstrates how ServiceNow can streamline and automate network service request handling using its low-code capabilities. By integrating the Service Catalog, Catalog Variables, UI Policies, Custom Tables, Flow Designer, Approval Management, and Email Notifications, the system provides a structured and efficient process for managing network requests from submission to completion.

Automation significantly reduces manual effort, minimizes processing delays, and ensures consistency throughout the request lifecycle. The implementation of approval workflows, automated record creation, and real-time notifications enhances transparency while maintaining complete audit trails and improving SLA compliance.

Overall, this project delivers a scalable, user-friendly, and reliable solution that improves operational efficiency, enhances the user experience, and provides a strong foundation for future enhancements and integration with advanced network automation tools.

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